$R_{\mu\nu} \stackrel{?}{=} K T_{\mu\nu}$ $\quad R_{\mu\nu} = R^{\rho}{}_{\mu g \nu}$

$\nabla^{\mu} T_{\mu\nu} = 0$ Energy Mom Conversation

$\nabla^{\mu} R_{\mu\nu} = \nabla^{\mu}(K T_{\mu\nu}) \Rightarrow \nabla^{\mu} R_{\mu\nu} = K \nabla^{\mu} T_{\mu\nu} = 0$

$\nabla^{\mu} R_{\mu\nu} \stackrel{?}{=} 0$ $\quad R_{\mu\nu} = R^{\rho}{}_{\mu g \nu}$

$R^{\rho}{}_{\mu\nu g} + R^{\rho}{}_{\nu g \mu} + R^{\rho}{}_{g \mu\nu}$

I. Bianchi Identity

$\nabla_{\rho} R_{\sigma\alpha\mu\nu} + \nabla_{\sigma} R_{\alpha\rho\mu\nu} + \nabla_{\alpha} R_{\rho\sigma\mu\nu}$

II. Bianchi Identity

$R_{\rho\lambda\alpha\beta} = -R_{\lambda\rho\alpha\beta}$

$= -R_{\alpha\beta\rho\lambda}$

$-R_{\rho\lambda\beta\alpha}$

I. Bianchi Identity

$\nabla_{\rho} R_{\sigma\alpha\mu\nu} + \nabla_{\sigma} R_{\alpha\rho\mu\nu} + \nabla_{\alpha} R_{\rho\sigma\mu\nu} = 0$

II. Bianchi Identity

$R_{\rho\lambda\alpha\beta} = -R_{\lambda\rho\alpha\beta}$

$= -R_{\alpha\beta\rho\lambda}$

$-R_{\rho\lambda\beta\alpha}$

$R_{\mu\nu} = K T_{\mu\nu} \Rightarrow \nabla^{\mu} R_{\mu\nu} \stackrel{?}{=} 0$

$g^{\sigma\mu} \nabla_{\rho} R_{\sigma\alpha\mu\nu} + g^{\sigma\mu} \nabla_{\sigma} R_{\alpha\rho\mu\nu} + g^{\sigma\mu} \nabla_{\alpha} R_{\rho\sigma\mu\nu} = 0$

$\nabla_{\rho} R^{\mu}{}_{\alpha\mu\nu} + \nabla^{\mu} R_{\alpha\rho\mu\nu} + \nabla_{\alpha} R_{\rho\nu} = 0$ $\quad -g^{\sigma\mu} \nabla_{\sigma} R_{\alpha\rho\mu\nu}$

$\nabla_{\rho} R_{\alpha\nu}$ $\quad \nabla^{\mu} \cdot -R_{\rho\alpha\mu\nu}$ $\quad \nabla^{\mu} \underbrace{R_{\alpha\rho\mu\nu}}_{R_{\rho\nu}}$

$\nabla^{\mu} \cdot -R_{\mu\nu\alpha\rho}$

$\nabla^{\mu} \cdot -R_{\alpha\rho\nu\mu}$

$$\Rightarrow \quad g^{\alpha\nu}\nabla_\rho R_{\alpha\nu} - \nabla^\mu R_{\rho\mu} - \nabla^\nu R_{\rho\nu} = 0$$

$$= \nabla_\rho R - \nabla^\mu R_{\rho\mu} - \nabla^\nu R_{\rho\nu} = 0 \qquad (\mu = \nu)$$

$$\nabla_\rho R - 2\nabla^\mu R_{\rho\mu} = 0$$

$$\Rightarrow \boxed{\nabla^\mu R_{\rho\mu} = \frac{1}{2}\nabla_\rho R}$$

$$\nabla^\mu R_{\mu\nu} = 0 = \nabla_\nu R \qquad \nabla_\mu R = 0 \Rightarrow R \text{ coef.}$$

$$g^{\mu\nu} R_{\mu\nu} = k T_{\mu\nu} g^{\mu\nu} \quad T$$

$$R = kT \qquad \nabla_\mu R = k\nabla_\mu T = 0$$

nice try but it doesnt true so $R_{\mu\nu} \neq kT_{\mu\nu}$

$$\boxed{\nabla^\mu R_{\rho\mu} = \frac{1}{2}\nabla_\rho R}$$

$$^\mu R_{\rho\mu} = \frac{1}{2}\nabla_\rho R \qquad R_{\mu\nu} \neq kT_{\mu\nu} \Rightarrow kT_{\mu\nu} = ?$$

$$\Rightarrow kT_{\mu\nu} \Rightarrow \nabla^\mu (kT_{\mu\nu}) \Rightarrow k\nabla^\mu T_{\mu\nu}$$

$$R_{\rho\mu} \Rightarrow \nabla^\mu (R_{\rho\mu}) = \frac{1}{2}\nabla_\rho R$$

$$\Rightarrow kT_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu}\nabla^\nu R$$

$$L[\gamma] = \int_\gamma (\mathcal{E}^{1/2}(x(\sigma))\, d\sigma \Rightarrow$$

$$\delta L[\gamma] = \int_\gamma \delta(\underbrace{\mathcal{E}^{1/2}(x(\sigma))\, d\sigma}_{ds} = 0$$

$$\Rightarrow \delta \mathcal{E}^{1/2} = \frac{1}{2}\mathcal{E}^{-1/2} \cdot \delta\mathcal{E} \Rightarrow \mathcal{E} = g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} \Rightarrow$$

$$\delta\mathcal{E}^{1/2} = \frac{1}{2}\mathcal{E}^{-1/2}\, \delta g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} + g_{\mu\nu}\left(\delta\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} + \frac{dx^\mu}{d\sigma}\frac{\delta dx^\nu}{d\sigma}\right) \qquad \mu = \nu \Rightarrow$$

$$\Rightarrow \frac{1}{2}\mathcal{E}^{-1/2}\left(\underbrace{\delta g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}}_{(i)} + \underbrace{2\, g_{\mu\nu}\frac{d\delta x^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}}_{(ii)}\right)$$

$$(ii)\ 2 g_{\mu\nu}\frac{d\delta x^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} \Rightarrow 2\{g_{\mu\nu}\frac{d}{d\sigma}\left(\frac{dx^\mu}{d\sigma}\delta x^\nu\right) = g_{\mu\nu}\{\left(\frac{dx^\mu}{d\sigma}\right)^2\delta x^\nu + \frac{dx^\mu}{d\sigma}\frac{d\delta x^\nu}{d\sigma}\}2$$

$$\Rightarrow \frac{dx^\mu}{d\sigma}\frac{d\delta x^\nu}{d\sigma} = \frac{d}{d\sigma}\left(\cancel{\frac{dx^\mu}{d\sigma}\delta x^\nu}\right) - \left(\frac{dx^\mu}{d\sigma}\right)^2\delta x^\nu \quad --- (ii)$$

$$\int_{\sigma_1}^{\sigma_2} 0$$

$$\Rightarrow (i) = 2\delta g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} = \delta x^\nu \cdot \partial_\alpha g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} \Rightarrow$$

$$\int +\frac{1}{2}\mathcal{E}^{-1/2}\left(2\partial_\alpha g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\alpha}{d\sigma} - \left(\frac{dx^\mu}{d\sigma}\right)^2\right)\delta x^\nu = 0 = \delta L[\gamma]$$

$$\Rightarrow \{\int\left(\mathcal{E}^{-1/2} g_{\mu\nu}\left(\frac{dx^\mu}{d\sigma}\right)^2 - \mathcal{E}^{-1/2}\frac{1}{2}\partial_\alpha g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}\right)\delta x^\nu = 0\}\times\left(-\mathcal{E}^{-1/2}\right)\} \qquad \boxed{\mathcal{E}^{+1/2} \cdot d\sigma = ds}$$

$$= \int\{g_{\mu\nu}\left(\frac{dx^\mu}{ds}\right)^2 - \frac{1}{2}\partial_\alpha g_{\mu\nu}\frac{dx^\mu}{ds}\frac{dx^\nu}{ds}\}\delta x^\nu = 0$$

Geodesic eq = $g_{\mu\nu}\left(\frac{dx^\mu}{ds}\right)^2 - \partial_\alpha g_{\mu\nu}\frac{dx^\mu}{ds}\frac{dx^\nu}{ds} = 0$ //

or

Geodesic eq = $\left(\frac{dx^\mu}{ds}\right)^2 + \frac{1}{2}\Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{ds}\frac{dx^\nu}{ds} = 0$ //

$$\Gamma^\alpha_{\mu\nu} =; \frac{1}{2} g^{\alpha\sigma}\left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}\right) //$$

$$ds^2\Big|_{\text{Newtonian}} = \underbrace{\left(1+\frac{2\phi}{c^2}\right)}_{g_{00}} c^2 (dt)^2 - \left(1-\frac{2\phi}{c^2}\right)(dx^2+dy^2+dz^2) = ds^2$$

$$g_{ij} = -\left(1-\frac{2\phi}{c^2}\right)\delta_{ij}$$

$$\frac{1}{ds^2} \Rightarrow \quad 1 = \left(1+\frac{2\phi}{c^2}\right)c^2\left(\frac{dt^2}{ds^2}\right) - \left(1-\frac{2\phi}{c^2}\right)\left(\frac{dx^2}{ds^2}+\frac{dy^2}{ds^2}+\frac{dz^2}{ds^2}\right)$$

(i) $\left(1+\frac{2\phi}{c^2}\right)c^2\left(\frac{dt^2}{ds^2}\right) = 1 + \left(1-\frac{2\phi}{c^2}\right)\left(\frac{dx^2}{ds^2}+\frac{dy^2}{ds^2}+\frac{dz^2}{ds^2}\right)$ int eq

$$\nu=\rho=0 \Rightarrow \frac{d}{ds}\left(g_{0\nu}\frac{dx^\nu}{ds}\right)=0 \Rightarrow \frac{d}{ds}\left(g_{00}\frac{dx^0}{ds}\right)=0$$

(ii) $\frac{d}{ds}\left[\left(1+\frac{2\phi}{c^2}\right)\frac{d(ct)}{ds}\right]=0$

(iii) $\delta_{ij}\frac{d}{ds}\left(1-\frac{2\phi}{c^2}\right)\frac{dx^j}{ds} = \frac{1}{2}\partial_i\left(1+\frac{2\phi}{c^2}\right)\left(\frac{d(ct)}{ds}\right)^2 - \frac{1}{2}\delta_{kl}\partial_i\left(1-\frac{2\phi}{c^2}\right) - \frac{1}{2}\partial_i\left(1-\frac{2\phi}{c^2}\right)\left(\frac{dx^2}{ds^2}+\dots\right)$

$$\rho = i = 1,2,3$$

$\frac{\phi}{c^2}\to 0 \quad \frac{v^2}{c^2}\to 0$ Weak Field/Slow motion approximation

$$\frac{d}{ds}\left[\left[1+\frac{2\phi}{c^2}\right]\frac{d(ct)}{ds}\right]=0 \Rightarrow \left(1+\underbrace{\cancel{\frac{2\phi}{c^2}}}_{0}\right)\frac{d(ct)}{ds} = a \Rightarrow \frac{d(ct)}{ds} = 1 \Rightarrow \frac{d(ct)}{a\,ds}=1$$

$$s' \to s$$

$$\Rightarrow \frac{dct}{ds} = 1 \Rightarrow$$

$$s = c(t+t_0)$$

$$s = c(t+t_0)$$

$$\left(\frac{dx^2}{ds^2}+\frac{dy^2}{ds^2}+\frac{dz^2}{ds^2}\right)$$

$$\left(1-\underbrace{\cancel{\frac{2\phi}{c^2}}}_{0}\right)c^2\underbrace{\left(\frac{dt}{ds}\right)^2}_{1} = 1 + \left(1-\underbrace{\cancel{\frac{2\phi}{c^2}}}_{0}\right)\left(\frac{v^2}{c^2}\right) \Rightarrow 1 = 1 \quad ///$$

$$* \quad \delta_{ij}\frac{d}{ds}\left(1-\cancel{\frac{2\phi}{c^2}}\right)\frac{dx^j}{ds} = \left(\frac{1}{2}\partial_i\left(\cancel{1}+\frac{2\phi}{c^2}\right)\left(\frac{dct}{ds}\right)^2 - \frac{1}{2}\partial_i\left(1-\frac{2\phi}{c^2}\right)\underbrace{\cancel{\left(\frac{dx^2}{ds^2}+\frac{dy^2}{ds^2}+\frac{dz^2}{ds^2}\right)}}_{0}\right.$$

$$\frac{d}{ds}\left(1-\cancel{\frac{2\phi}{c^2}}\right)\frac{dx^j}{ds} = \frac{1}{2}\partial_i\frac{2\phi}{c^2}\frac{(-c^2)dt^2}{ds^2}$$

$$\Rightarrow \frac{d^2x^j}{dt^2} = \partial_i\phi \Rightarrow \left\{\frac{d^2x^j}{dt^2} = -\partial^j\phi\right\}\vec{e}_j \Rightarrow \quad m\frac{d^2(x^j e_j)}{dt^2} = -\nabla\phi\, m$$

$$U$$

$g_{\mu\nu} = \underset{\text{minkw}}{\eta_{\mu\nu}} + h_{\mu\nu}$ perturbation effect

$\eta_{\mu\nu} = g_{\mu\nu} - h_{\mu\nu}$ $\quad g_{\mu\nu} g^{\nu\lambda} = \delta_\mu^\lambda$ (Kronecker weitrass Delta)

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \Rightarrow g_{\mu\nu} g^{\nu\lambda} = (\eta_{\mu\nu} + h_{\mu\nu})(\eta^{\nu\lambda} - h^{\nu\lambda})$

$= \eta_{\mu\nu}\eta^{\nu\lambda} - \cancel{\eta_{\mu\nu} h^{\nu\lambda}} + \cancel{h_{\mu\nu}\eta^{\nu\lambda}} - \underbrace{h_{\mu\nu} h^{\nu\lambda}}_{\text{neglect}} = \delta_\mu^\lambda$

$\eta^{\mu\alpha} h_{\alpha B} = (g^{\mu\alpha} - h^{\mu\alpha}) h_{\alpha B} =$

$= g^{\mu\alpha} h_{\alpha B} - \underbrace{\cancel{h^{\mu\alpha} h_{\alpha B}}}_{\text{neglect}}$

$= h^\mu_B \Rightarrow \eta^{\mu\alpha} h_{\alpha B} = h^\mu_B$

christoffel;

$\Gamma^\mu_{\mu\nu} = \frac{1}{2} g^{\mu\lambda} \cdot [\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\nu} - \partial_\lambda g_{\mu\nu}]$

$\Gamma^\mu_{00} = \frac{1}{2} g \quad (\cancel{\partial_0 g_{\lambda 0}}^{0} + \cancel{\partial_0 g_{\lambda 0}}^{0} - \partial_\lambda g_{00})$

$\Rightarrow \Gamma^\mu_{00} \cong -\frac{1}{2} g^{\mu\lambda} \partial_\lambda g_{00} \qquad g^{\mu\lambda} = \eta^{\mu\lambda} - h^{\mu\lambda} \qquad g_{\mu\lambda} = \eta_{\mu\lambda} + h_{\mu\lambda}$

$\Rightarrow \frac{-1}{2}(\eta^{\mu\lambda} - h^{\mu\lambda}) \partial_\lambda g_{00} \Rightarrow -\frac{1}{2}(\eta^{\mu\lambda} - h^{\mu\lambda}) \partial_\lambda (\underbrace{\eta_{00}}_{-1} + h_{00})$

$-\frac{1}{2}\eta^{\mu\lambda} \underbrace{\cancel{\partial_\lambda (-1)}}_{0} - \frac{1}{2}\eta^{\mu\lambda}\partial_\lambda h_{00} + \frac{1}{2} h^{\mu\lambda} \cdot \underbrace{\cancel{\partial_\lambda (-1)}}_{0} + \frac{1}{2} h^{\mu\lambda} \cancel{\partial_\lambda h_{00}}$

$\Gamma^\mu_{00} = \frac{-1}{2}\eta^{\mu\lambda}\partial_\lambda h_{00} \Rightarrow \Gamma^0_{00} \cong 0 \qquad \Gamma^\mu_{00} = -\frac{1}{2}\eta^{\mu\lambda}\partial_\lambda h_{00}$

Geodesic eq.

(I) $\frac{d^2 x^\mu}{d\tau^2} - \frac{1}{2}\partial_\alpha g_{\mu\nu} \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \cong 0$

(II) $\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha B}\left(\frac{dx^\alpha}{d\tau}\right)^2 = 0 \Rightarrow$

$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00}\left(\frac{dx^0}{d\tau}\right)^2 \cong 0 \quad (\alpha = 0)$

$\frac{d^2 x^0}{d\tau^2} = 0 \quad (\mu = 0) \qquad \frac{dx^0}{d\tau} = sbt$

$\frac{dx^{\mu 2}}{d\tau^2} + \Gamma^\mu_{00}\left(\frac{dx^\alpha}{d\tau}\right)^2 \cong 0 \qquad \mu = \alpha \qquad \Gamma^\mu_{00} = -\frac{1}{2}\eta^{\mu\lambda}\partial_\lambda h_{00} \qquad \mu = \lambda \Rightarrow$

$\Gamma^\mu_{00} = -\frac{1}{2}\partial_\lambda h_{00}$

$$\frac{dx^{\mu 2}}{d\tau^2} + \Gamma^{\mu}_{00}\left(\frac{dx^0}{d\tau}\right)^2 \simeq 0 \qquad \Gamma^{\mu}_{00} = -\frac{1}{2}\partial_{\mu} h_{00}$$

$$\frac{dx^{\mu 2}}{d\tau^2} - \frac{1}{2}\partial_{\mu} h_{00}\left(\frac{dx^0}{d\tau}\right)^2 \simeq 0 \Rightarrow \frac{dx^{\mu 2}}{d\tau^2} = \frac{1}{2}\partial_{\mu} h_{00}\left(\frac{dx^0}{d\tau}\right)^2$$

$$\Rightarrow \left(\frac{d\tau}{dx^0}\right)^2 \frac{d^2x^{\mu}}{d\tau^2} = \frac{1}{2}\partial_{\mu} h_{00}$$

$$\Rightarrow \frac{d^2x^i}{dt^2} \simeq \frac{1}{2}\partial_i h_{00}$$ Geodesic Newtonian Limit

$$\Rightarrow \frac{d^2x^i}{dt^2} = -\partial_i \Phi \quad \Rightarrow h_{00} = -2\Phi$$

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \Rightarrow g_{00} = \underbrace{\eta_{00}}_{-1} + h_{00} = -1 - 2\Phi = -(1+2\Phi)$$

$$g_{00} \simeq -(1+2\Phi) \qquad \Phi = -\frac{GM}{r}$$